PROBLEM 39

Particle on a String with a Plate

Original statement · Problems.pdf, p. 27

Plate Edge Contact

A heavy point particle P with mass m is connected with point O on the upper surface of a thin square plate by a massless, inextensible string of length ℓ. The distance between O and the edge of the plate is d. The string hangs over the edge of the plate and can slide without friction. You can assume that the string remains tense and in contact with the plate at all times. We denote the geometric point where the string is in contact with the plate edge as Q. We define the inertial reference frame $S_0 : \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$. We define the angle θ between the Ox_0 axis and the string segment OQ, and the angle ϕ between a vertical plane containing the plate edge and the line perpendicular to the plate edge that passes through P. Initially at time $t = 0, \theta(0) = \phi(0) = \pi/3$ rad, and the modulus of the velocity of P is $v_0^P(0) = \sqrt{\ell g}$, where g is the magnitude of the acceleration due to gravity.

(Hint: Note that under the conditions of the problem, the two angles marked with dashed lines are equal).

(a) How many degrees of freedom does the system have?

(b) Obtain the general expression of a unit vector in the direction from Q to P as a function of θ and ϕ.

(c) Obtain the general expression for the position, velocity, and acceleration vectors of Q and P in S_0 as a function of θ, ϕ, their derivatives, and the parameters of the problem.

(d) Identify and express mathematically all the forces acting on P.

(e)

Find a set of second order differential equations to determine the evolution of θ and ϕ in time, free of any other unknowns.

(f) Does the tension force exert work on P? Provide proof of your answer.

(g) Derive the general expression of the mechanical energy of P in S_0 and compute its value initially. Is this a conserved quantity of motion? Justify your answer.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle P of mass m is suspended by an inextensible massless string of length ℓ anchored at point O on a horizontal thin square plate. The string passes over the straight edge of the plate (at perpendicular distance d from O) at contact point Q.
- **CDOFs:** 2 degrees of freedom: the angle θ defining the position of contact point Q along the edge, and the swing angle ϕ of the hanging segment QP.
- **Initial State ($t = 0$):** $\theta(0) = \phi(0) = \pi/3$ rad, with $|\mathbf{v}_P(0)| = \sqrt{\ell g}$.
- **Parameters:** $m, \ell, d, g > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Frame S_0 : $\{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$. Plate in horizontal plane $z_0 = 0$.
- Distance from O to edge point Q: $r_Q = \frac{d}{\cos\theta}$.
- Hanging segment length: $\rho = \ell - r_Q = \ell - \frac{d}{\cos\theta}$.
- **Geometry assumption (read from the figure, Problems.pdf PDF page 27).** The plate edge is the straight line $x_0 = d$, parallel to $\mathbf{j}_0$ (so Ox_0 is perpendicular to the edge); θ is the angle between Ox_0 and OQ, and ϕ is the angle between the vertical and the line through P perpendicular to the edge, which lies in a plane normal to the edge. The statement text alone does not fix the direction of the edge, so this reading relies on the figure.

- Unit vector $\mathbf{e}_{QP}$ pointing from Q to P. The edge is frictionless, so the two string segments make equal angles with it (the hint of the statement): the unit vector from Q to O is $\mathbf{e}_{QO} = -(\cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0)$, hence $\mathbf{e}_{QP}\cdot\mathbf{j}_0 = -\mathbf{e}_{QO}\cdot\mathbf{j}_0 = \sin\theta$. The remaining part of $\mathbf{e}_{QP}$ lies in the plane normal to the edge, along $\sin\phi\,\mathbf{i}_0 - \cos\phi\,\mathbf{k}_0$, with magnitude $\sqrt{1-\sin^2\theta} = \cos\theta$:

$$\mathbf{e}_{QP} = \cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0, \qquad |\mathbf{e}_{QP}|^2 = \cos^2\theta(\sin^2\phi+\cos^2\phi) + \sin^2\theta = 1$$

- **Swing basis.** The hanging segment QP swings in the plane normal to the edge, i.e. about the edge direction $\mathbf{j}_0$. The basis $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_0, \mathbf{k}_1\}$ is the inertial basis rotated by ϕ about $\mathbf{j}_0$, with $\mathbf{i}_1 = \cos\phi\,\mathbf{i}_0 + \sin\phi\,\mathbf{k}_0$ and $\mathbf{k}_1 = -\sin\phi\,\mathbf{i}_0 + \cos\phi\,\mathbf{k}_0$; the columns are the components in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\phi & 0 & -\sin\phi \\ 0 & 1 & 0 \\ \sin\phi & 0 & \cos\phi \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\phi + \sin^2\phi = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- In this basis the unit vector of Phase 2 reads $\mathbf{e}_{QP} = \sin\theta\,\mathbf{j}_0 - \cos\theta\,\mathbf{k}_1$, since $-\mathbf{k}_1 = \sin\phi\,\mathbf{i}_0 - \cos\phi\,\mathbf{k}_0$ is the direction of the line through P perpendicular to the edge. Both the matrix properties and the identity with the expression of $\mathbf{e}_{QP}$ in $\mathcal{B}_0$ were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a)-(c): Kinematics of Q and P

Position of contact point Q:

$$\mathbf{r}_Q = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0$$

Velocity of Q:

$$\mathbf{v}_Q = \frac{d\dot{\theta}}{\cos^2\theta}\,\mathbf{j}_0$$

Position of particle P:

$$\mathbf{r}_P = \mathbf{r}_Q + \rho\,\mathbf{e}_{QP} = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0 + \left(\ell - \frac{d}{\cos\theta}\right)(\cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0)$$

Velocity $\mathbf{v}_P$ is found by differentiating $\mathbf{r}_P(t)$ with respect to time.

Part (d): Forces on P

1. Gravity: $\mathbf{F}_g = -mg\mathbf{k}_0$.

2. String tension: $\mathbf{T} = -T\mathbf{e}_{QP}$.

Part (e): Equations of Motion

Projecting Newton's second law along the independent tangent directions yields two coupled second-order ODEs for $\theta(t)$ and $\phi(t)$.

Part (f): Work Done by Tension on P

> **Pedagogical Justification:** The tension acts along $-\mathbf{e}_{QP}$ on P, so its power is $-T\,\mathbf{e}_{QP}\cdot\mathbf{v}_P$. The string is inextensible, so the hanging length $\rho = \ell - d/\cos\theta$ changes only because Q slides along the edge (direction $\mathbf{j}_0$).

Differentiate $\mathbf{r}_P = \mathbf{r}_Q + \rho\,\mathbf{e}_{QP}$ (product rule): $\mathbf{v}_P = \mathbf{v}_Q + \dot{\rho}\,\mathbf{e}_{QP} + \rho\,\dot{\mathbf{e}}_{QP}$. Since $|\mathbf{e}_{QP}| = 1$, $\mathbf{e}_{QP}\cdot\dot{\mathbf{e}}_{QP} = \frac{1}{2}\frac{d}{dt}|\mathbf{e}_{QP}|^2 = 0$, so

$$\mathbf{e}_{QP}\cdot\mathbf{v}_P = \mathbf{e}_{QP}\cdot\mathbf{v}_Q + \dot{\rho}$$

$\mathbf{v}_Q = \dfrac{d\dot{\theta}}{\cos^2\theta}\,\mathbf{j}_0$ has only a $\mathbf{j}_0$ component, so $\mathbf{e}_{QP}\cdot\mathbf{v}_Q = \sin\theta\,\dfrac{d\dot{\theta}}{\cos^2\theta}$. With the chain rule $\frac{d}{dt}\left(\frac{1}{\cos\theta}\right) = \frac{\sin\theta}{\cos^2\theta}\dot{\theta}$, the rate of change of the hanging length is $\dot{\rho} = -\dfrac{d\dot{\theta}\sin\theta}{\cos^2\theta}$. Therefore

$$\mathbf{e}_{QP}\cdot\mathbf{v}_P=\frac{d\dot\theta\sin\theta}{\cos^2\theta}-\frac{d\dot\theta\sin\theta}{\cos^2\theta}=0\implies\mathbf{T}\cdot\mathbf{v}_P=-T\,\mathbf{e}_{QP}\cdot\mathbf{v}_P=0$$

The tension therefore does no work on P ($W_T=0$), in agreement with the official key (Problems.pdf, PDF page 76).

Part (g): Mechanical Energy Conservation

Since the plate is stationary (scleronomic constraint), the tension does no work (part (f)) and gravity is conservative, so E is conserved. With z_0 upwards, $z_P=-\rho\cos\theta\cos\phi=-(\ell\cos\theta-d)\cos\phi$ and the gravitational potential energy is $+mgz_P$:

$$E=\frac{1}{2}m|\mathbf{v}_P|^2+mgz_P=\frac{1}{2}m|\mathbf{v}_P|^2-mg(\ell\cos\theta-d)\cos\phi=\text{constant}$$

At $t=0$: $|\mathbf{v}_P|^2=\ell g$, $\cos\theta=\cos\phi=\frac{1}{2}$, so

$$E_0=\frac{1}{2}m\ell g-mg\left(\frac{\ell}{2}-d\right)\frac{1}{2}=\frac{mg\ell}{4}+\frac{mgd}{2}$$

Parts (d) and (e) are only outlined here: the explicit coupled differential equations for θ and ϕ are not derived in this note.

PHASE 4

Physical Interpretation & Dimensional Verification

- The system acts as a variable-length spherical pendulum where string length changes deterministically as Q slides along the boundary.
- **Dimensional check:** $[\rho]=[\ell-d/\cos\theta]=\text{m}$; $[mg\ell]=[mgd]=\text{J}$, so $E_0=mg\ell/4+mgd/2$ is an energy; $[\mathbf{e}_{QP}]=1$.
- **Limit $d\to0$:** the edge passes through O, so $Q=O$, $\rho=\ell$ and $\mathbf{e}_{QP}=(\cos\theta\sin\phi,\ \sin\theta,\ -\cos\theta\cos\phi)$ is the unit vector of a spherical pendulum of length ℓ whose angle γ with the downward vertical satisfies $\cos\gamma=\cos\theta\cos\phi$. The energy reduces to $E_0=mg\ell/4$, which equals $\frac{1}{2}m\ell g-mg\ell\cos\theta\cos\phi=\frac{1}{2}m\ell g-mg\ell/4$ at $\theta=\phi=\pi/3$.

- **Numerical check ($m = 1$ kg, $\ell = 2$ m, $d = 0.5$ m, $g = 9.81\ \mathrm{m/s^2}$):** $E_0 = mg\ell/4 + mgd/2 = 7.3575$ J, and directly $\tfrac{1}{2}m\ell g + mgz_P(0) = 9.81 - 2.4525 = 7.3575$ J with $z_P(0) = -(\ell/2 - d)/2 = -0.25$ m.

Source: Problems.pdf (Page 27)